

The Solar Cycle – Teacher Copy

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In this class, students identify patterns in data from Google Trends and the Trove archive, geomagnetic data and sunspot data to determine the length of the solar cycle.

You can either provide the ready-made data plots to students (see **SolarCyclePlots**) or, for more advanced students, you can remove the plots from the worksheet and task students with downloading and making their own plots in Microsoft Excel or in a Jupyter notebook, following the guidance in the document '**SolarCycleDataLocations**'.

Australian Curriculum: Science F-10 Version 9.0 links:

Analyse and connect a variety of data and information to identify and explain patterns, trends, relationships and anomalies (AC9S9I05 AC9S10I05).

Pre-requisite knowledge:

- Students should have a conceptual understanding of the Earth's magnetic field, geomagnetic activity and the aurora.
- Students should be familiar with time series data and be able to identify trends, measure time between peaks and compare multiple variables.
- Students should have a basic understanding that nature has natural cycles and that the length of these can be calculated as the time between peaks.

Aims:

- Identify patterns in social trend data and geomagnetic data to determine the length of the solar cycle.
 - Understand the link between sunspots, the geomagnetic field and auroral activity and why we use geomagnetic indices.
 - Understand how social trends can reflect underlying natural cycles.
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Part 1: Google Trends data

Navigate to Google Trends: <https://trends.google.com/trends/>

What is Google Trends?

[Google Trends](#) is a free public tool that allows you to explore trends in Google searches for specific terms over time and in different areas of the world (Raffoul, 2024). It produces graphs and data tables that you can download and analyse. Data is presented on a scale of 0 – 100, where 100 is the peak interest in searches.

1. Type 'aurora australis' into the search bar
2. Change the location to 'Worldwide'
3. Change the time to '2004 - present' (Google began collecting trends data in 2004)
4. Leave the search type as 'Web Search'

Your search should look like **Figure A**.

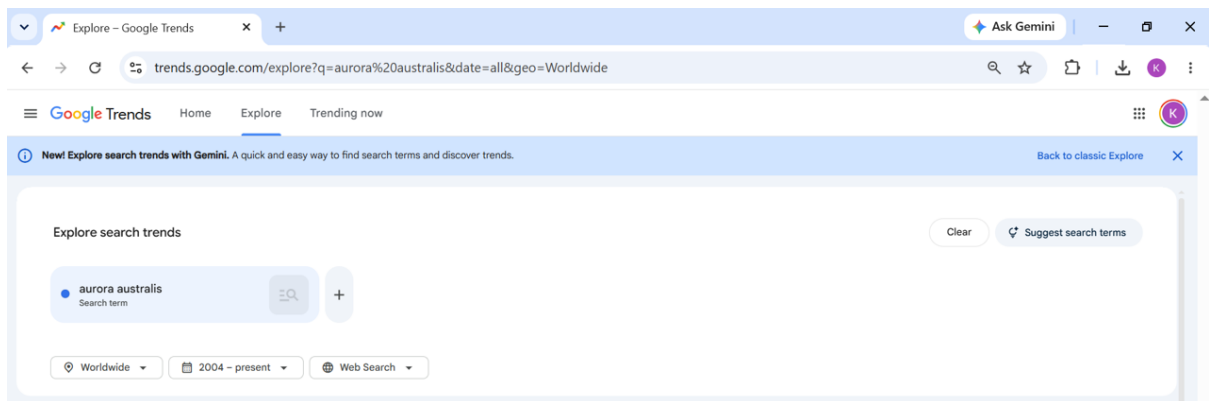


Figure A: Google Trends search for 'aurora australis' worldwide from 2004 – present (Google Trends, 2026).

A line graph will appear below your search criteria. Explore the graph by moving your mouse across the graph – you can click on points along the graph to see specific dates.

Your graph should look like **Figure 1** below – this figure is the same Google Trends data plotted using Python.

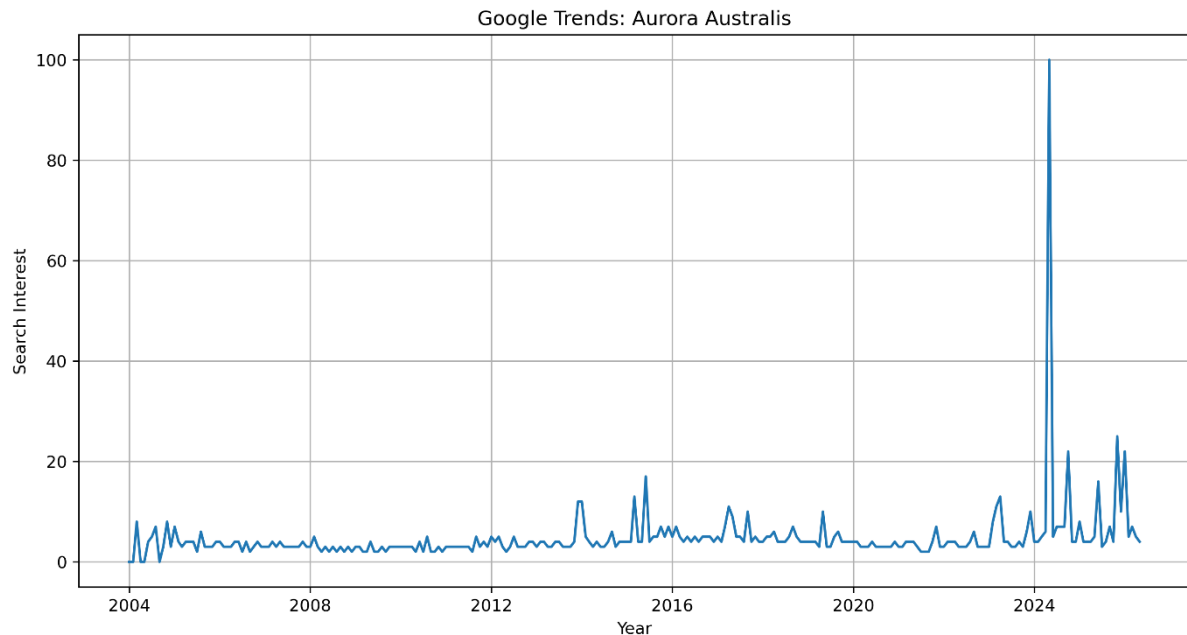


Figure 1: Worldwide search interest for ‘Aurora Australis’ (Jan 2004 – June 2026). Data source: Google Trends, 2026

(<https://trends.google.com/explore?q=aurora%2520australis&date=all&geo=Worldwide>)

Question 1: Why do you think we are searching for ‘aurora australis’ instead of just ‘aurora’ or ‘aurora borealis’?

If we simply searched for ‘aurora’, we would be including lots of irrelevant Google search data into our search – think the Disney princess Aurora, the singer AURORA and even energy companies called Aurora! You can test this for yourself by just googling the word ‘aurora’ and looking at the results.

‘Aurora borealis’ refers to auroras in the northern hemisphere. We are concerned with southern hemisphere/ Australian auroras, aka the ‘aurora australis’, in this class.

Question 2: Is there a particular point on your graph where there is a big spike (i.e. peak interest) in Google searches for ‘aurora australis’? When was this? What do you think happened during this month?

Peak interest in Google searches for the term ‘aurora australis’ occurred in **May 2024**.

In May 2024, the most extreme geomagnetic storm of solar cycle 25, termed the ‘Gannon Storm’, occurred (Lawrence et al., 2025). This storm triggered widespread and low latitude auroras that could even be seen as far south as southern England!

The auroral event was so globally widespread that Google Searches for ‘aurora australis’ surged to an all-time peak, with lots of interested people wanting to find out how to see the aurora for themselves or just see photos and news articles about the event.

Some resources relating to this event are linked below:

Hansen, K. (2024) *Historic geomagnetic storm dazzles*. NASA Earth Observatory, 16 May. Available at: <https://science.nasa.gov/earth/earth-observatory/historic-geomagnetic-storm-dazzles-152815/>

Jackson, D. (2024) *Mancs stunned as Northern Lights glow above Greater Manchester in rare solar storm*. The Manc, 11 May. Available at: <https://themanc.com/news/mancs-stunned-as-northern-lights-glow-above-greater-manchester-in-rare-solar-storm/>

Lawrence, E. et al. (2025). The geomagnetic and geoelectric response to the May 2024 geomagnetic storm in the United Kingdom, *Frontiers in Astronomy and Space Science*, vol. 12, 1550923. <https://doi.org/10.3389/fspas.2025.1550923>

Figure 2 below is the same plot but zoomed in to remove the big spike you just identified. This big spike is an outlier (a value far outside of the rest of the dataset) and makes trends in the rest of the data difficult to see.

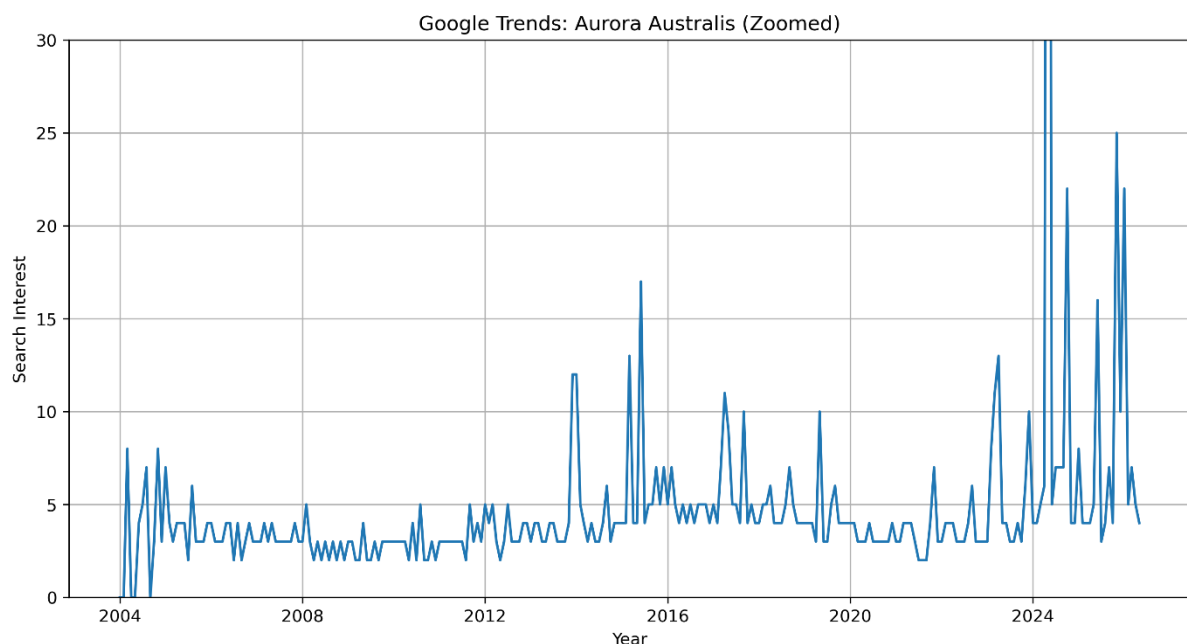


Figure 2: Zoomed in Worldwide search interest for ‘Aurora Australis’ (Jan 2004 – June 2026). Data source: Google Trends, 2026

(<https://trends.google.com/explore?q=aurora%2520australis&date=all&geo=Worldwide>)

Question 3: Can you identify any overall patterns on your graph i.e. search interest increasing to a peak and then decreasing back down? If you can, mark the boundaries on to your worksheet.

Students should aim to mark on the boundaries of solar cycle 24 (2008–2019) (SILSO) onto their graph (**Figure B**).

There is a weakly visible trend of search interest increasing from 2008 to a peak around 2014 which corresponds to the peak of this solar cycle and then searches fall back down towards 2019 (the solar minima) (SILSO).

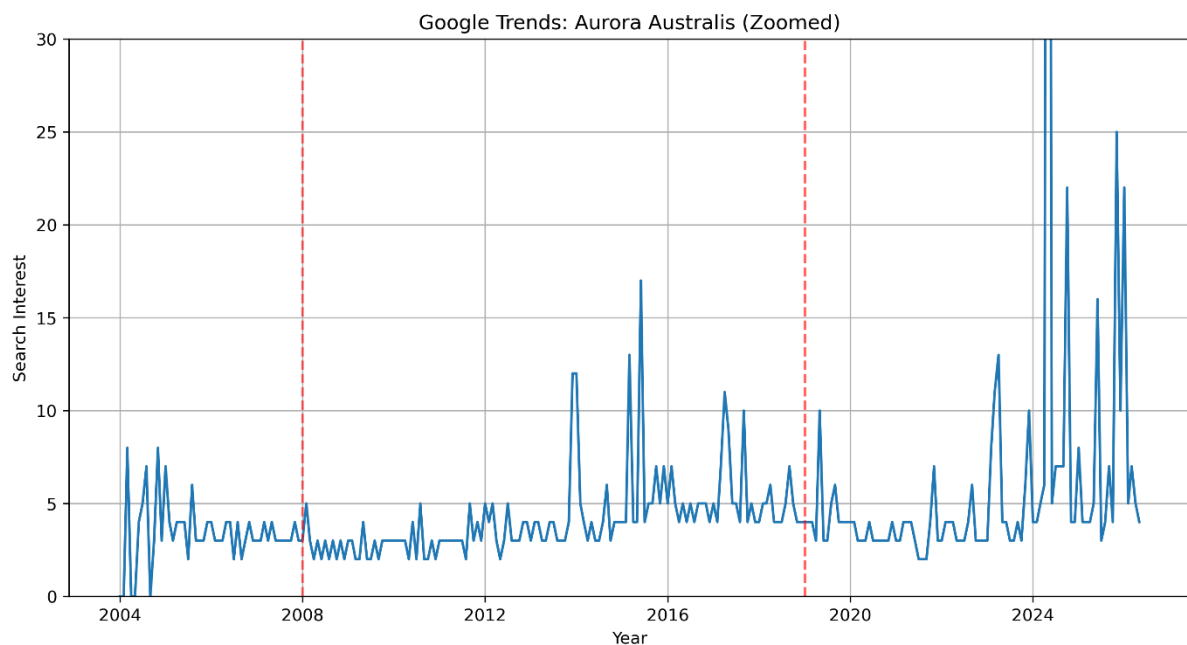


Figure B: Zoomed in Google Trends data for ‘Aurora Australis’ worldwide from 2004 – June 2026. Data source: Google Trends, 2026
(<https://trends.google.com/explore?q=aurora%2520australis&date=all&geo=Worldwide>)

Red lines mark the boundaries of solar cycle 24 (2008-2019) (SILSO).

If students cannot identify this at this early stage in the class (which is to be expected), they should come back to this figure after studying the next datasets and attempt see the trend then.

Do not worry if you cannot identify any convincing trends at this stage.

Is 2004-2026 long enough to see any trends? We will now consider social trends data from the Trove archive over a longer time period and see if that helps us.

Part 2: Trove archive data

Figure 3 below shows data from number of references to ‘Aurora Australis’ each year, according to the Trove archive of Australian newspapers.

The data is from 1822 to 2023 (Note that the Trove archive is limited by copyright restrictions – anything less than 70 years old is restricted for most sources, hence the counts drop away after 1956).

What is Trove?

[Trove](#) is an online platform from the National Library of Australia (NLA) and contains more than 14 billion digital items such as digitised newspapers, books, maps and photographs. Trove Content on the Trove database is fully searchable, making it an excellent research tool for all things Australian (ARDC, 2025; National Library of Australia).

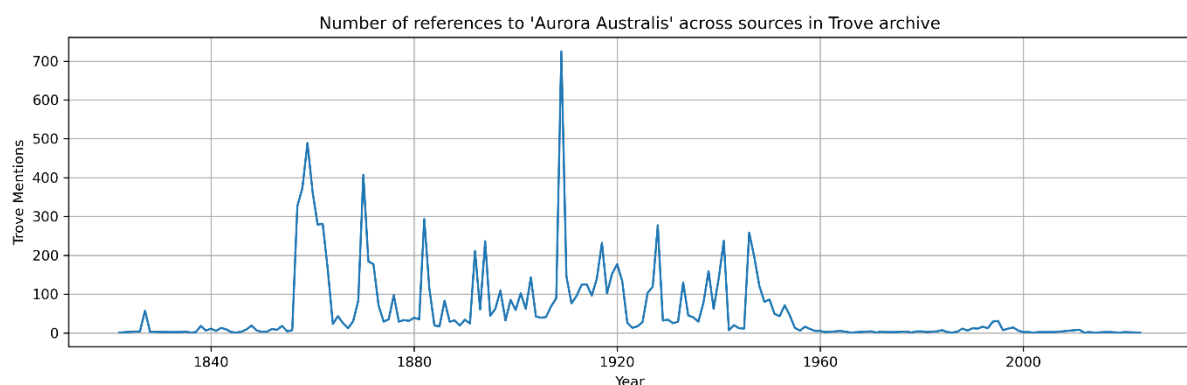


Figure 3: Number of references to “Aurora Australis” across sources in Trove archive (1822-2023), compiled from National Library of Australia Trove database.

Question 4: Can you identify any cyclical/ repeating trends in this data? If so, how long do cycles last? Mark the boundaries of any cycles you identify on your worksheet.

Hint: Try identifying peaks in Trove Mentions and measuring the amount of time between consecutive peaks?

Students should see that there are regular repeating cycles where Trove mentions rise to a maximum and fall back down to a minimum.

The length of these cycles is approximately 11 years. Students should try to mark these on their copy of the graph (see **Figure C**).

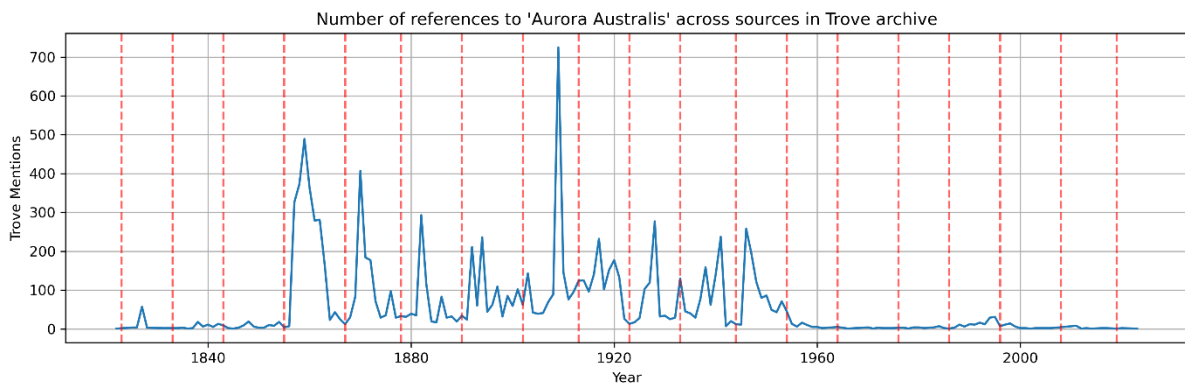


Figure C: Number of references to “Aurora Australis” across sources in Trove archive (1822-2023), compiled from National Library of Australia Trove database.

Red lines mark the boundaries of 11-year solar cycles (SILSO).

Part 3: Geomagnetic observatory data

Geomagnetic observatories are permanent recording stations that measure changes in the Earth’s magnetic field over time. There are geomagnetic observatories all around the world, and these are connected as a global network (INTERMAGNET).

Figure 4 is a graph of raw data from the Canberra geomagnetic observatory from 1991 – present. The data is for the x component, which shows geomagnetic activity in the magnetic north direction.

INSERT FIGURE – needs to be created by downloading and plotting Canberra geomagnetic x component data from 1991 – present from https://imag-data.bgs.ac.uk/GIN_V1/GINForms2

This could also be printed out as a large paper copy (multiple sheets taped together) and students could look at it from eye level with the table (similar to seismic interpretation) to try and see trends.

Figure 4: Geomagnetic activity (x component) from the Canberra Geomagnetic Observatory 1991 – the present. Data source: INTERMAGNET data portal

Question 5: Can you identify any patterns in this data over time? If not, why do you think patterns are hard to see?

Students should find that it is very difficult to identify regular repeating trends/ cycles in the raw geomagnetic observatory data. They should note that the data is noisy and difficult to interpret.

They should be visually overwhelmed by the plot – this then leads into the teaching point in **Part 4** about why we need geomagnetic indices.

Part 4: Indexed geomagnetic data

As you saw in part 3, the raw data collected at geomagnetic observatories (like the observatory in Canberra) is often complex and patterns are difficult to see.

Geoscientists use geomagnetic indices to solve this problem (Geoscience Australia). When indices are applied to raw data, noise is reduced and trends become easier to see.

In the field of geomagnetism, two important indices are the k index and the aa index.

- The k index measures geomagnetic activity every 3 hours relative to a quiet day on a scale of 0-9 for a specific observatory (Geoscience Australia).
- The aa index is an average measure of geomagnetic activity that uses the k index values from two observatories on opposite sides of the world – Canberra, Australia and Hartland, England (Geoscience Australia).

Figure 5 shows the yearly averaged aa index from 01 January 1868 to 01 June 2026. The monthly mean values are in blue, and the yearly average is in orange.

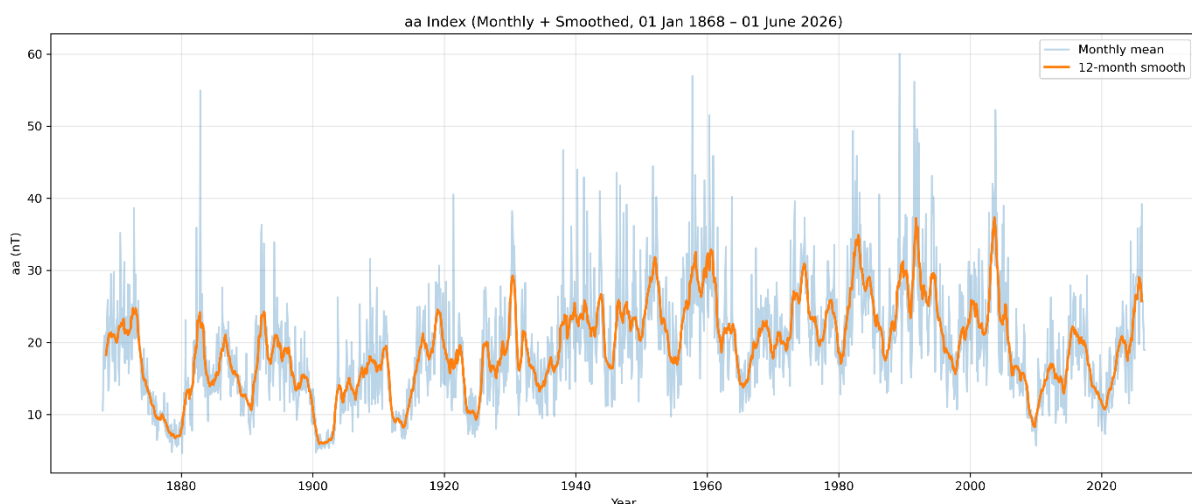


Figure 5: Geomagnetic aa index, monthly average (blue) and yearly smoothed (orange).

Data source: Mayaud, P.-N., Menvielle, M., & Chambodut, A. (2023). aa geomagnetic index. EOST. (Dataset). <https://doi.org/10.25577/9z05-v751>

Question 6: Do you notice any repeating patterns in the aa index data?

Similar to the Trove data, students should see that there is a cyclical pattern in aa value rising to a maximum and then falling back down to a minimum.

Question 7: Where are the peaks in geomagnetic activity? Label these and calculate approximately how many years there are between them.

Students should mark the peaks in aa value onto their copy of the graph and see that there is approximately 11 years between them (see **Figure D**).

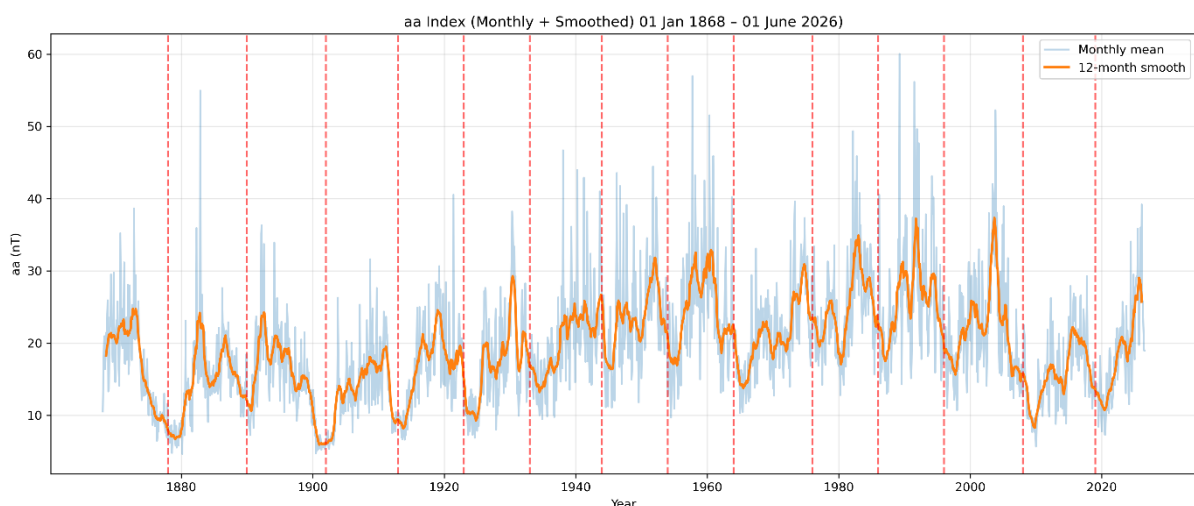


Figure D: Geomagnetic aa index, monthly average (blue) and yearly smoothed (orange).

Data source: Mayaud, P.-N., Menvielle, M., & Chambodut, A. (2023). aa geomagnetic index. EOST. (Dataset). <https://doi.org/10.25577/9z05-v751>

Red lines mark the boundaries of 11-year solar cycles ([SILSO](#)).

Part 5: Sunspot numbers

We will now consider patterns in the number of sunspots over time.

Sunspots are dark, cooler regions on the surface of the Sun caused by intense magnetic activity (NASA, 2025). They are linked to solar eruptions, which drive geomagnetic activity on Earth. The number of sunspots varies over time.

Figure 6 shows how the number of sunspots has changed between 1770 to 2026.

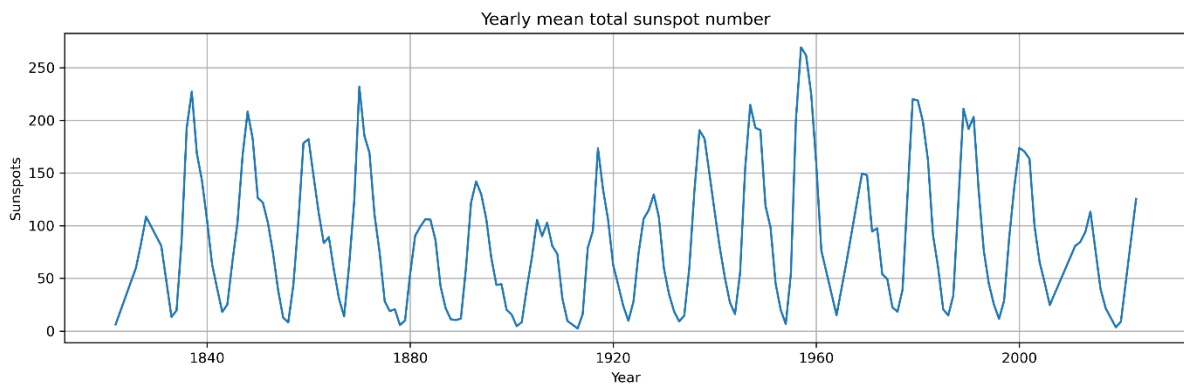


Figure 6: Yearly mean total sunspot number (1770-June 2026). Data source: WDC-SILSO, Royal Observatory of Belgium, Brussels, DOI: <https://doi.org/10.24414/qnza-ac80>

Question 8: Do you notice any repeating patterns in the sunspot data?

The 11-year cycle in sunspot data should be obvious to students in this plot.

Question 9: Where are the peaks in sunspot number? Label these and calculate approximately how many years there are between them. These peaks are known as **solar maxima**.

Students should mark the solar maxima (sunspot number peaks, these are listed by SILSO [here](#)) and calculate that there is 11 years between them. See **Figure E**.

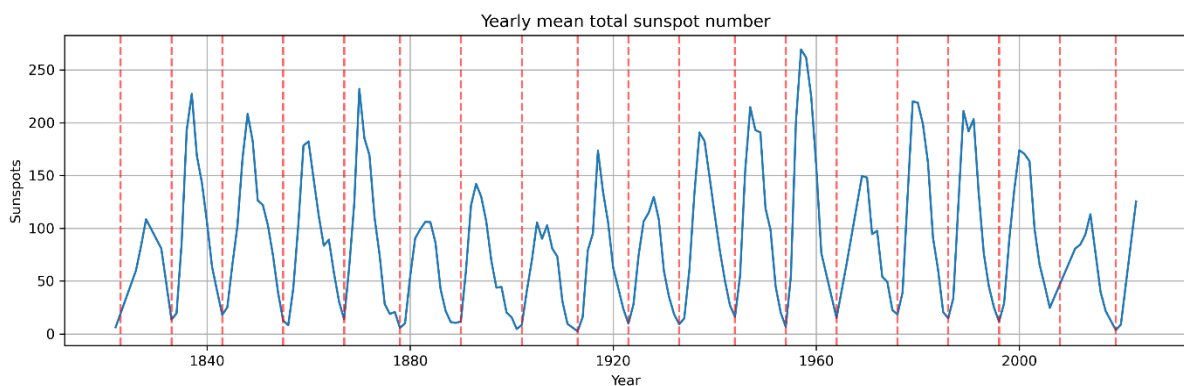


Figure E: Yearly mean total sunspot number (1770-June 2026). Data source: WDC-SILSO, Royal Observatory of Belgium, Brussels, DOI: <https://doi.org/10.24414/qnza-ac80>

Red lines mark the boundaries of 11-year solar cycles ([SILSO](#)).

Part 6: Summary

From your study of all the datasets, answer the following questions:

Question 10: You should have identified a regular repeating cycle in the Trove, aa and sunspot data. The length of this cycle should be approximately the same. How many years does one cycle take?

11 years

This is the duration of the **solar cycle**. The solar cycle is defined by a **solar maximum** in the middle of the cycle and a **solar minimum** at the cycle's start/ end (NASA, 2025).

Question 11: Is auroral activity stronger at the solar maximum or the solar minimum? Why do you think this?

The below figure (**Figure 7**) of the yearly mean total sunspot number and Trove archive 'aurora australis' mentions on top of one another may help you.

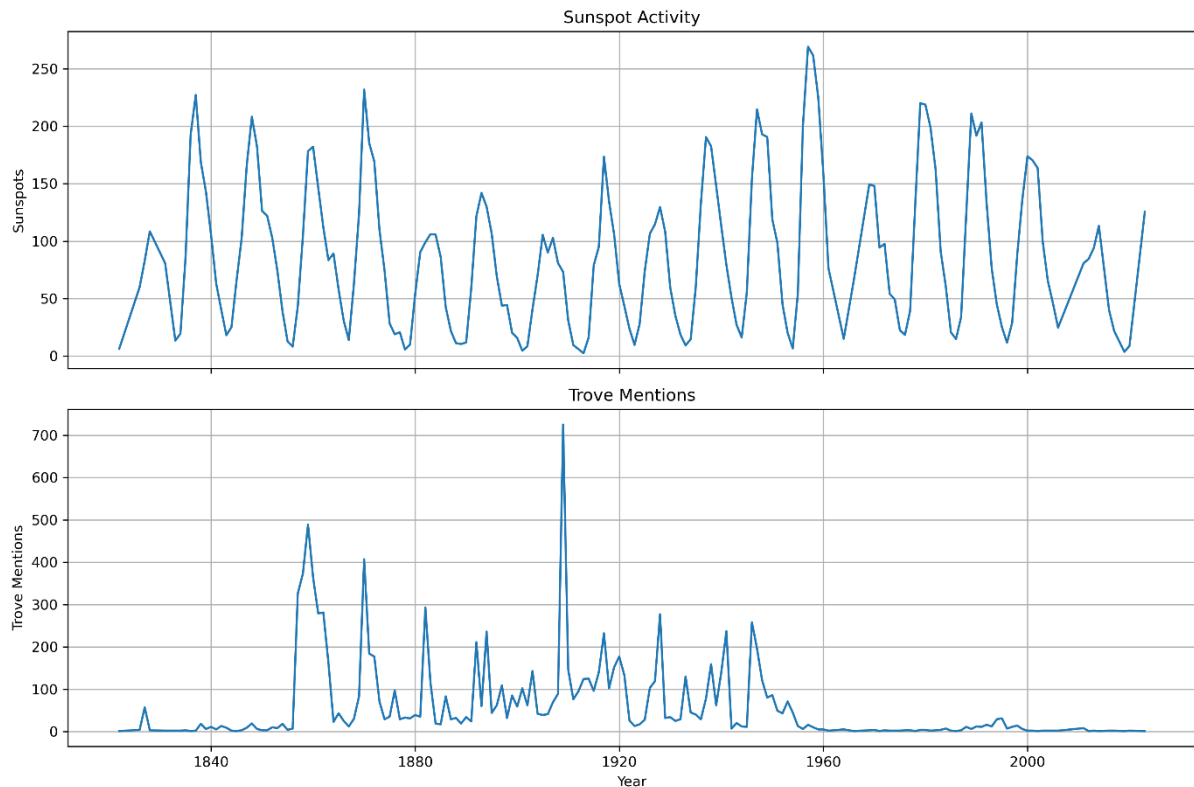


Figure 7: Yearly mean total sunspot number (1770-June 2026) (WDC-SILSO, Royal Observatory of Belgium, Brussels, DOI: <https://doi.org/10.24414/qnza-ac80>) above Number of references to “Aurora Australis” across sources in Trove archive (1822-2023), compiled from National Library of Australia Trove database.

The Trove archive data can be taken as a proxy for auroral activity. Students should conclude that at solar (sunspot) maxima there is more auroral activity (evidenced by an increase in Trove mentions to aurora australis).

The Trove archive data and sunspot trends follow each other closely and the solar cycle boundaries can be used to define cycles in both datasets. This relationship is shown in **Figure F**.

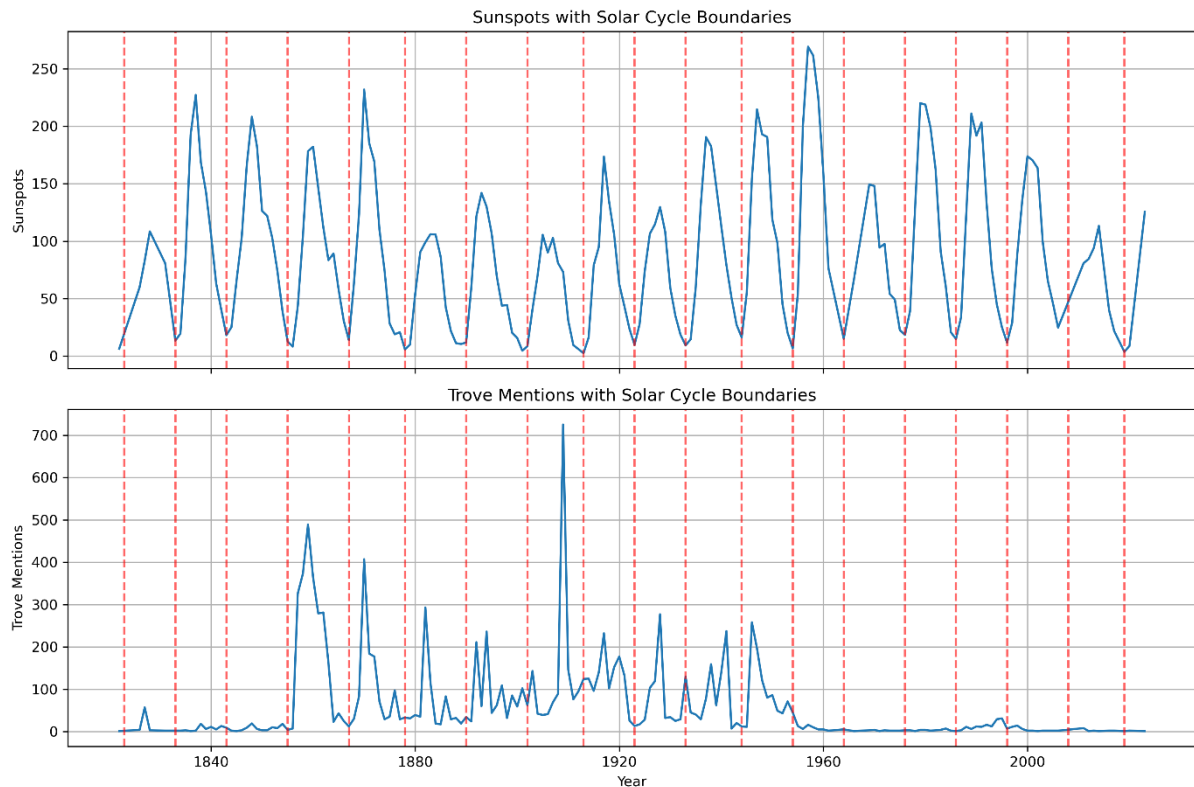


Figure F: Yearly mean total sunspot number (1770-June 2026) (WDC-SILSO, Royal Observatory of Belgium, Brussels, DOI: <https://doi.org/10.24414/qnza-ac80>) above Number of references to “Aurora Australis” across sources in Trove archive (1822-2023), compiled from National Library of Australia Trove database.

Red lines mark the boundaries of 11-year solar cycles ([SILSO](#)).

Question 12: Why do you think the social trend data from Google Trends and the Trove archive closely resemble the solar cycle?

There is no right answer to this question. Students should speculate that auroras spark human interest so the number of Google searches/ news articles referencing the aurora increase when there is an increase in auroral activity at a solar maximum and fall towards solar minima where auroras are less frequent.

References:

Australian Research Data Commons (ARDC) (2025). *Trove for Researchers*. Available at: <https://ardc.edu.au/resource/trove-for-researchers/>

Geoscience Australia. *Geomagnetic Indices: aa index*. Available at: <https://geomagnetism.ga.gov.au/geomagnetic-indices/aa-index>

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<https://www.sidc.be/SILSO/datafiles>

Sunspot Index and Long-term Solar Observations (SILSO). *Solar Cycles Min/Max*.

Available at: <https://sidc.be/SILSO/cyclesminmax>

WDC-SILSO, Royal Observatory of Belgium, Brussels, DOI:

<https://doi.org/10.24414/qnza-ac80>

This resource was developed with assistance from ChatGPT (OpenAI, 2026), used to generate and refine instructional ideas.